EBU6475 Microprocessor Systems Design

Assessment Guidance: GenAI-CDIO Authentic Assessment with Challenge-Based Learning (Version 1)

By Jonathan Loo

1. Purpose of this Guidance

This guidance explains how the assessment framework works—what it evaluates, why it is structured this way, and how you can engage with it confidently and effectively. The framework mirrors professional engineering practice, where the engineering process is paramount: how problems are framed, decisions are justified, knowledge gaps are addressed, and ideas are tested and refined.

The final system evidence that process. A working solution demonstrates how requirements were interpreted, how design choices were implemented, how components were integrated, and how behaviour was validated in practice.

The assessment framework expects clear evidence of a systematic, defensible engineering process that leads to a coherent outcome.

2. What the Assessment Evaluates

This assessment evaluates your development as an engineer through the demonstration of **Engineering Competence (EC)** within an **Authentic Assessment (AA)** context, learning through engagement with a real-world, challenging, and open-ended engineering problem that reflects the complexity and uncertainty of professional practice.

Within this framework, your EC is demonstrated through the application of a systematic engineering design process and its realisation in a functioning system.

The engineering process is assessed through how effectively you:

- understand and frame the engineering problem,
- make reasoned and defensible design decisions,
- identify and address knowledge and skills gaps,
- implement and integrate hardware and software solution,
- test and validate system behaviour,
- reflect on outcomes and improve iteratively.

The system product is assessed through how clearly it demonstrates:

- accurate interpretation of requirements,
- justified translation of design decisions into implementation,
- effective integration of components under real constraints,
- validated and reliable system behaviour.

System performance is further evidenced through structured benchmarking under standardised conditions, providing objective, comparable results for how well and how reliably the system meets defined criteria.

This framework aligns with key principles of **AA**, including contextual realism, complexity, integration of technical and professional knowledge, higher-order thinking, collaboration, iteration, validated performance, responsible use of digital tools, and professional accountability, and maps to the UK Engineering Council **AHEP4** competence areas (C1–C18), including:

- Knowledge and problem analysis (C1–C3)
- Design and engineering judgement (C4–C6)
- Ethics and professional responsibility (C7–C11)
- Experimentation and evaluation (C12–C14)
- Communication and teamwork (C16–C17)
- Independent professional development (C18)

3. How You Engage with the Assessment Lifecycle Process

You engage with the assessment challenge through the GenAI–CDIO workflow shown in Figure 1. This structured lifecycle provides the formal process through which EC is developed, evidenced, and demonstrated.

You progress through each stage:

- **Conceive:** define the problem by clarifying requirements, constraints, and knowledge gaps.
- **Design:** develop and compare solutions, evaluate trade-offs, and justify a defensible approach.
- **Implement:** translating the design into hardware/software solution, integrating components under real constraints, and refining the system through testing and iteration.
- **Operate:** validate system behaviour, benchmark performance under standardised conditions, analyse results, and improve based on evidence.

The stages are interconnected rather than strictly sequential. As new insights or performance results emerge, earlier decisions may be revisited and refined. Iteration reflects professional engineering practice and strengthens design coherence.

Throughout the lifecycle, reasoning should remain explicit. Decisions, validation activities, test results, and refinements are expected to be clearly documented. The assessment therefore evaluates not only the final system, but the quality of justification, validation, and improvement over time.

GenAI may support knowledge exploration, comparison of alternative solutions, code drafting, troubleshooting, and performance analysis. However, it does not replace engineering judgement. Responsibility for decisions, validation, and outcome integrity remains with you. All AI-supported outputs should be critically evaluated and verified through testing.

Finally, during the Operate stage, structured benchmarking provides objective and comparable evidence of system performance. When combined with documented reasoning and validated results, this evidence demonstrates the EC described in Section 2.

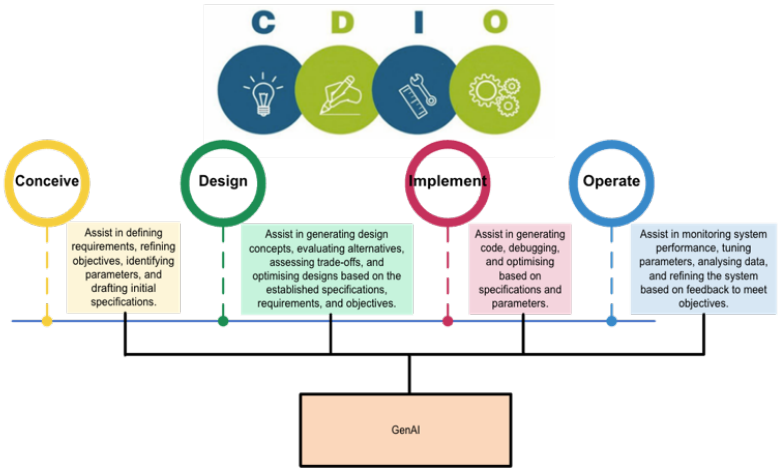

Fig. 1 GenAI-CDIO workflow

4. How You Use GenAI Responsibly

Building on the CDIO engagement described in Section 3, you may use GenAI as a structured support tool throughout the engineering process. It can assist with:

- learning unfamiliar concepts,
- exploring and comparing design ideas,
- reasoning about implementation strategies,
- debugging, optimisation, and performance analysis.

However, responsible use requires critical evaluation. GenAI can support your thinking, but it does not replace engineering judgement. You remain fully accountable for design decisions, validation activities, documentation, and the integrity of the final system.

When using GenAI, your reasoning must stay transparent. You should be able to explain:

- what questions you asked,
- how GenAI influenced your reasoning,
- what you accepted, modified, or rejected,
- how you verified outcomes through testing or from observed system behaviour.

In professional engineering practice, digital tools are expected to enhance reasoning, not obscure it. Your documentation should therefore make your decision-making process clear, traceable, and defensible.

Effective use of GenAI is demonstrated not by relying on automated outputs, but by the quality of your validation, justification, and reflection applied to those outputs. Responsible and transparent use of digital tools is a key aspect of modern engineering professionalism and directly supports your development as an engineer.

5. Making Your Engineering Reasoning Visible

As you progress through the CDIO framework and use GenAI responsibly, it is essential that your engineering reasoning remains explicit, transparent, and easy to follow. The assessment evaluates not only the final system you produce but also the clarity, logic, and defensibility of the decisions that led to it.

Your documentation should demonstrate:

- how your ideas developed across the CDIO stages,
- how and why your design decisions evolved as new information emerged,
- how the final implementation reflects those decisions,
- how testing, benchmarking, and validation informed further refinement.

Effective documentation makes your decision pathway clear, coherent, and well-justified. The quality of reasoning matters far more than the quantity of material you submit. The goal is to provide evidence of:

- well-supported decisions,
- validated results,
- disciplined iteration and systematic refinement.

Acceptable forms of evidence include design rationales within your documentation, development logs, version histories, test results, performance data, and reflective commentary on alternative approaches or unsuccessful attempts.

Maintaining transparent reasoning strengthens the link between your engineering process, the final product, and the competence you demonstrate through your work.

6. Collaborative Learning and Teamwork

Engineering practice is inherently collaborative, and this assessment reflects that reality through structured teamwork. Within your team, you are expected to practise professional engineering behaviours, including:

- discuss and challenge ideas constructively,
- distribute responsibilities appropriately,
- integrate individual contributions into a coherent system,
- communicate technical reasoning clearly to peers.

Collaborative learning strengthens engineering competence by exposing assumptions, refining reasoning, and improving design quality through dialogue and shared problem-solving. Although the system is developed collectively, each student's engineering competence is assessed through the visibility, quality, and justification of their individual contributions within the team context.

Effective teamwork is not measured by simple agreement, but by constructive engineering discussion, responsible task ownership, and shared accountability for outcomes. Strong teams demonstrate open communication, clear coordination, and collective commitment to producing a coherent and well-validated system.

7. Reflection and Professional Growth

Reflection is a core engineering skill. Engineers continually evaluate how their understanding evolves, how challenges are addressed, and how systems can be improved. Effective reflection makes the reasoning behind decisions visible and strengthens both technical and professional growth.

Reflecting on decisions, trade-offs, limitations, and opportunities for future improvement demonstrates professional maturity and readiness for continued development. It shows that you can critically assess your work, recognise why certain choices were made, and identify how your system could be strengthened.

Learning through imperfection is an essential part of authentic engineering practice. A thoughtfully analysed system that clearly explains its limitations and outlines evidence-based improvements can demonstrate strong engineering competence, even when performance is not ideal.

Ultimately, reflection supports growth and learning, it is not about fault-finding, but about understanding, improving, and developing as an engineer.

8. Professional Practice and Integrity

Professional engineering practice demands integrity, transparency, and accountability. This assessment reflects those expectations. Building on reflective practice and responsible use of engineering tools, you are required to demonstrate clear, evidence-based decision-making and honest documentation throughout your work.

Transparency in GenAI use, justified trade-offs, and accurate reporting of system behaviour are essential components of responsible engineering practice. Students are expected to demonstrate integrity by:

- disclosing and validating all GenAI-assisted reasoning,
- ensuring technical decisions are defensible and supported by evidence,
- documenting processes truthfully and accurately,
- maintaining clear records of testing procedures and system behaviour,
- upholding integrity in both individual and team contributions.

Your work should be sufficiently clear and traceable that another engineer can follow your reasoning, understand your design decisions, and reproduce your validation process independently.

Engineering integrity means that decisions are supported by genuine evidence, testing results are reported accurately, limitations are acknowledged honestly, and system behaviour is described faithfully—even when performance is imperfect.

These practices prepare you for professional environments where accountability, transparency, and engineering responsibility are fundamental. Demonstrating integrity is not simply about avoiding misconduct; it is about building trust in your engineering judgement and in the systems you design.

9. Key Messages to Remember

- Engineering competence develops through a structured, disciplined, and well-justified design process.
- The CDIO lifecycle provides clarity and guidance for navigating open-ended engineering challenges.
- The engineering process drives the quality of the final product; the product, in turn, demonstrates your reasoning and design decisions.
- GenAI can support exploration and idea development, but engineering judgement, validation, and accountability remain your responsibility.
- Clear evidence, transparent documentation, and systematic benchmarking make your engineering competence visible.
- Effective teamwork enhances design quality and reflects authentic professional practice.
- Iteration, analysis, and thoughtful reflection contribute more to engineering development.

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